

Part A · Multiple Choice

2 points each

1. **d) Compression**

p. 200 — compression squeezes rocks together, causing them to fold or fracture. Tension dominates at divergent boundaries and shear at transform boundaries.

2. **a) plastic deformation**

p. 200 — deep in the crust, where temperatures and pressures are high, rocks tend to deform plastically. Near the surface, or under sudden stress, they break instead.

3. **b) an anticline**

p. 201 — a syncline is the reverse: it bends downward, putting the youngest rocks at the center. A monocline is a simple bend that leaves the layers no longer horizontal.

4. **a) the hanging wall drops down relative to the footwall**

p. 202 — normal faults occur under tensional stress and are often responsible for uplifting mountain ranges. In a reverse fault the footwall drops relative to the hanging wall.

5. **c) About 80%**

p. 203 — the basin is lined with convergent and transform boundaries. Another 15% occur in the Mediterranean-Asiatic Belt, and the remaining 5% elsewhere. Roughly 95% of all quakes occur at plate boundaries of some kind.

6. **b) 10**

p. 207 — both scales are logarithmic. The amplitude grows ten times per integer, while the energy released grows about thirty times.

Part B · True or False

1 point each

7. **FALSE**

That describes the epicenter. The focus (or hypocenter) is the initial point within the crust where the rupture begins. In about 75% of earthquakes it lies in the top 10–15 km. (p. 203)

8. **TRUE**

p. 209 — ground shaking itself seldom kills; falling structures and subsequent fires do. Damage depends less on magnitude than on the quality of construction and the ground beneath it.

Part C · Short Answer

3 points each

9. **Mercalli measures intensity — what people felt and the damage done — so it is qualitative and varies by location. Moment magnitude measures the total energy released. Most seismologists use moment magnitude because it most accurately reflects energy released and damage caused.**

Rubric: 1 pt Mercalli — describes earthquakes by what nearby residents felt and the damage to structures; qualitative, since it depends on building codes and on the rock beneath, not on energy released at the focus. 1 pt moment magnitude — calculated from the area of fault ruptured and the distance the ground moved, giving total energy released. 1 pt — moment magnitude has replaced the Richter scale and is what most seismologists use, because Richter can rate a single sharp jolt higher than a longer, more energetic quake. (pp. 206–208)

10. **Liquefaction is what happens when water-saturated sediments are shaken and begin behaving like quicksand. The Wasatch Front is built on loose soil left by Lake Bonneville with a water table near the surface.**

Rubric: 1 pt — sediments saturated with water lose strength during shaking and behave like quicksand; solid bedrock vibrates less and suffers less damage. 2 pts — much of the Wasatch Front is loose soil left over from Lake Bonneville, and near the Jordan River and its floodplain the water table sits close to the surface, so shaking drives water upward and oversaturates the soil, making it extremely unstable. Credit the 1906 San Francisco example, where much of the damage in low-lying areas was attributed to liquefaction. (pp. 209–210)